# EE337 - Microprocessor Lab Project

April 19, 2026

**Aim:**

The aim of this project is to understand and implement communication between an 8051 microcontroller and external devices using the I2C protocol. The goal is to develop a basic device driver that allows the microcontroller to interact with sensors like the BMP280 and modules like the RTC. Through this project, we aim to learn how to read and write data, display information on an LCD, and send data to a computer using serial communication. This helps in building a strong foundation in embedded systems and device driver development.

# 1 PHASE - 1

**Aim:**

To write a bare-metal program on the 8051 microcontroller to read the identification (WHO_AM_I) register of the BMP280 sensor using the I2C protocol, and display the intermediate and final values on the LCD for verification.

## 1.1 Code

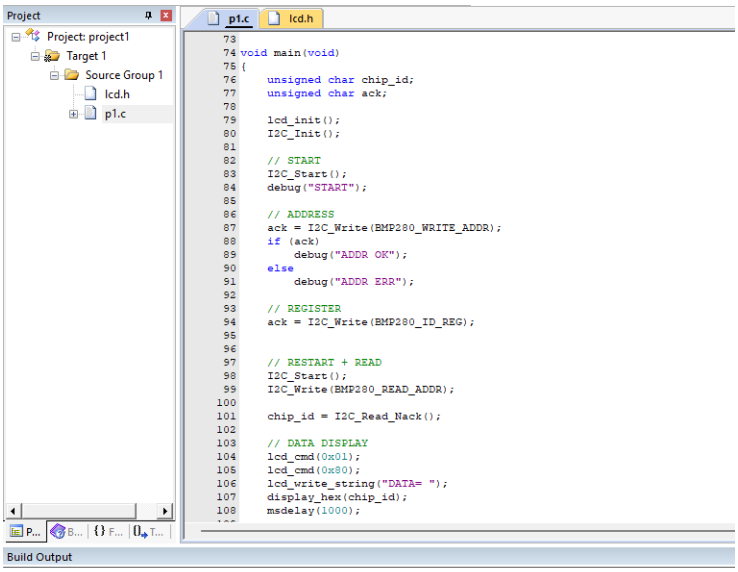



Figure 1: Phase 1 main.c

## 1.2 Observation

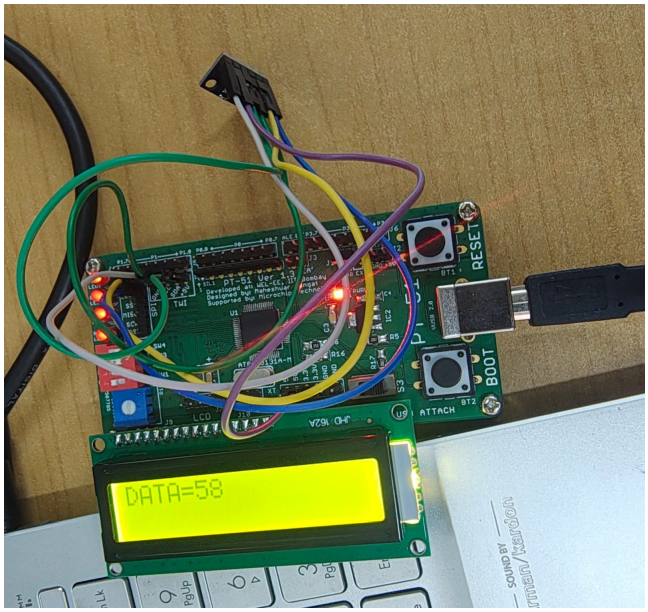



Figure 2: Phase 1 output

## 1.3 Conclusion

The I2C communication between the 8051 microcontroller and the BMP280 sensor was successfully established. The WHO_AM_I register was read correctly, and the chip ID value **0x58** was observed on the LCD, confirming proper device addressing and communication. Intermediate debugging messages such as START condition and address acknowledgement were also verified. This confirms that the I2C protocol was implemented correctly at the bare-metal level.

# 2 PHASE - 2

**Aim:**

To develop a custom I2C device driver library for the 8051 microcontroller that enables reading from and writing to any arbitrary I2C device memory address by implementing basic I2C functions such as Start, Stop, Read, Write.

## 2.1 Code

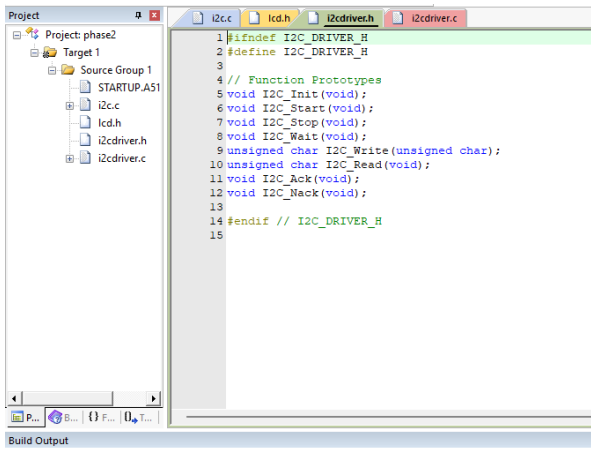



Figure 3: Phase -2 i2c.h

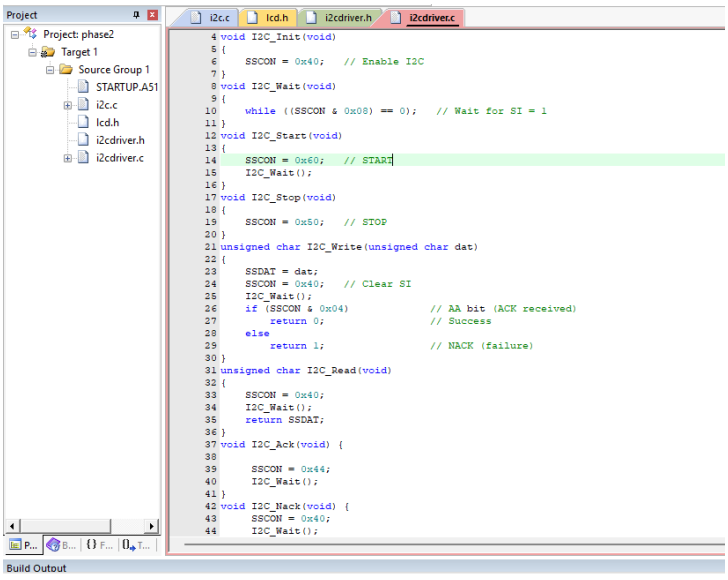



Figure 4: Phase -2 i2c.c

## 2.2 Observation

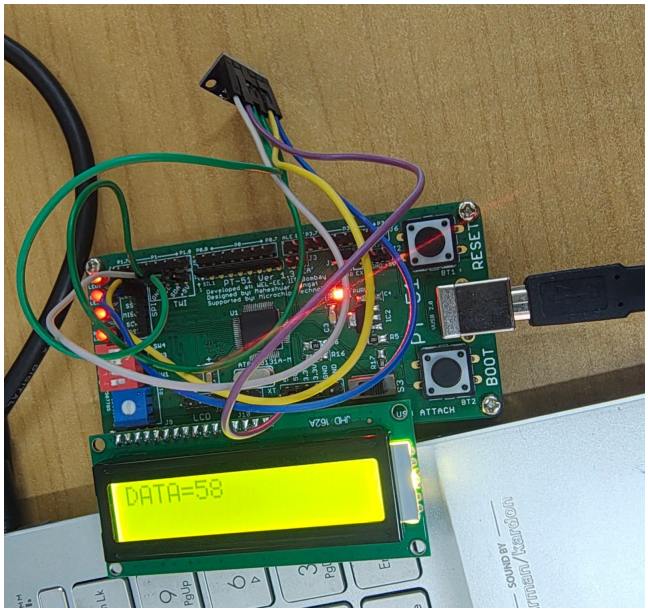



Figure 5: Phase 2 output

## 2.3 Conclusion

A custom I2C device driver was successfully developed and tested. The implemented functions such as Start, Stop, Read, Write, ACK, and NACK enabled reliable communication with the BMP280 sensor. Using this driver, the WHO_AM_I register was again read successfully, confirming correct abstraction of low-level operations.

# 3 PHASE - 3

**Aim:**

To interface an RTC module with the 8051 microcontroller using the developed I2C driver, read and process time data in BCD format, display the timestamp on the LCD, and transmit time and temperature data to a PC using UART communication.

## 3.1 Code

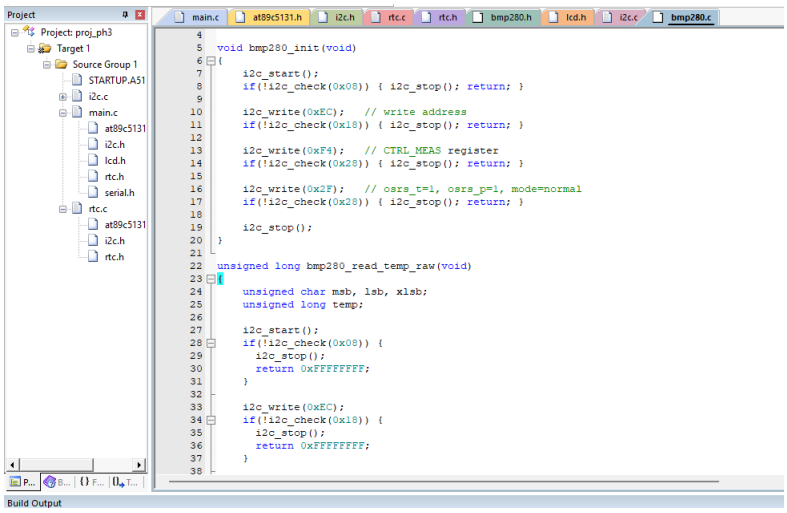



Figure 6: Phase -3 bmp.c

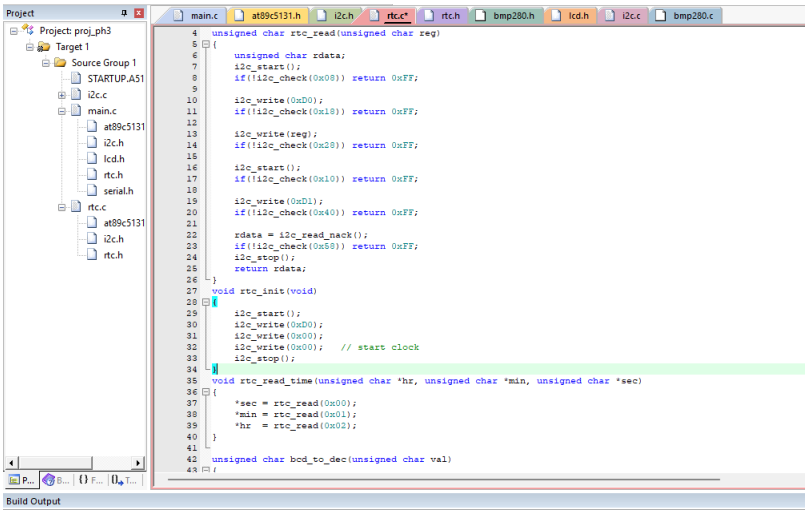



Figure 7: Phase -3 rtc.c

## 3.2 Observation

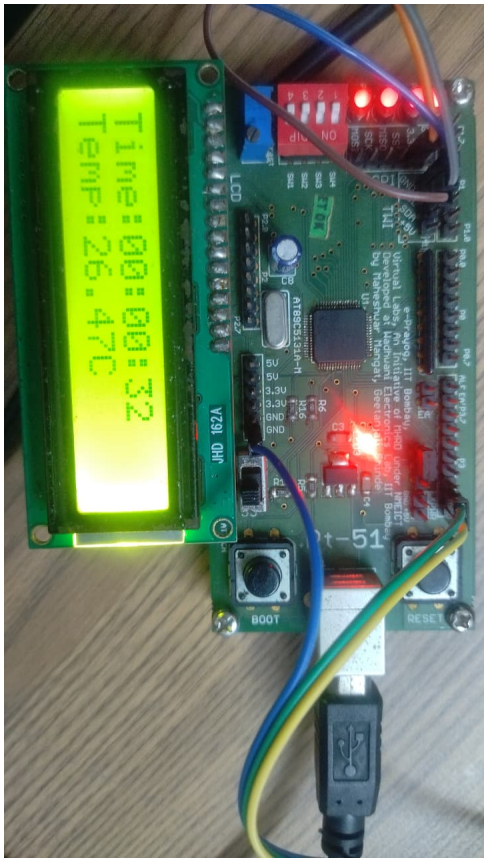



Figure 8: Phase 3 output




Figure 9: Phase 3 output

## 3.3 Conclusion

The RTC module was successfully interfaced using the developed I2C driver, and timer values were read correctly from the registers. The BCD data was converted to decimal and displayed on the LCD in the format **HH:MM:SS**.Additionally, the temperature from the BMP280 sensor was measured and found to be approximately **26°C**, after applying calibration and compensation formulas. The timer and temperature data were also successfully transmitted to the PC via UART, confirming proper integration of all subsystems.

# 4 Overall Conclusion

In this project, we successfully implemented I2C communication between the 8051 microcontroller and external devices. We verified communication by reading the WHO_AM_I register (**0x58**), developed a reusable I2C driver, and interfaced the RTC and BMP280 sensor. Timer and temperature data were correctly read, processed, displayed on the LCD, and transmitted via UART. This project provided a strong understanding of embedded communication, device drivers, and sensor interfacing.